

Prompt2Print: A Guardrailed AI Workflow for Typography-to-STL Generation

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Introduction & Motivation

Creating printable typographic reliefs usually requires CAD expertise. General-purpose AI 3D tools like meshy.ai are primarily designed for broad visual asset creation rather than exact typography and application-specific geometric validation. Prompt2Print preserves user input, explores a bounded parametric design space, and pairs every generated STL with an explicit validation report. The AI selects aesthetic parameters; deterministic software builds, validates, and reports each artifact in the workflow.

Main Workflow

- User Input.** The user enters a 1–8-letter word and selects a typographic style and supported base construction.
- AI Design Agent.** GPT-5.6-Terra API proposes four schema-constrained parameter sets within the fixed design space; it does not create geometry or assign verdicts.
- Proposal Preview.** The interface presents the four design proposals for comparison while preserving the user’s original text.
- Parameter Validation.** Deterministic rules verify the schema, parameter ranges, registered fonts, supported bases, and predicted feature constraints.
- Geometry Generation.** OpenSCAD measures the actual text outline, fits each design to the build envelope, and exports the candidate STLs.
- Validation & Checks.** Fixed geometry and feature gates evaluate every exported STL. Failed candidates enter the AI Repair Layer and are rechecked for at most two repair rounds.
- Output & Download.** The user selects a validated candidate for profile-specific slicing and downloads the STL with its SHA-256-bound report for 3D printing those work of art.

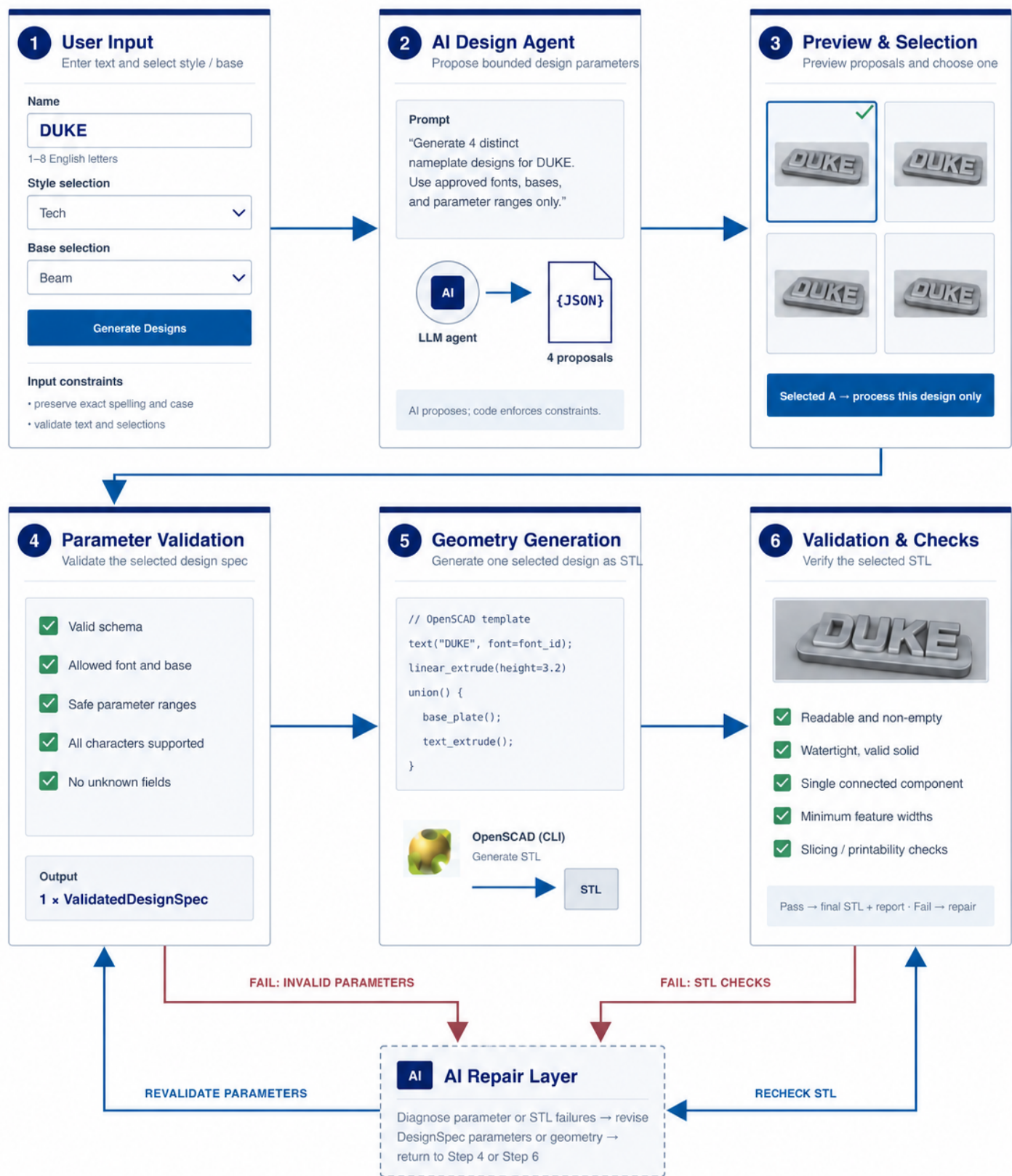


Figure 1. Workflow Chart

Design Outputs

Each request yields four distinct interpretations of the same unchanged text. A registered font supplies the skeleton; the AI varies weight, contrast, slant, corners, inline and hollow features, stripes, and per-letter rotation, shift, and scale. Designs may use plaque, text-contour, beam, wedge, or stand bases. Every preview is derived from the exported STL rather than redrawn artwork.



Figure 2. Letter Forms with Standing Bases Preview

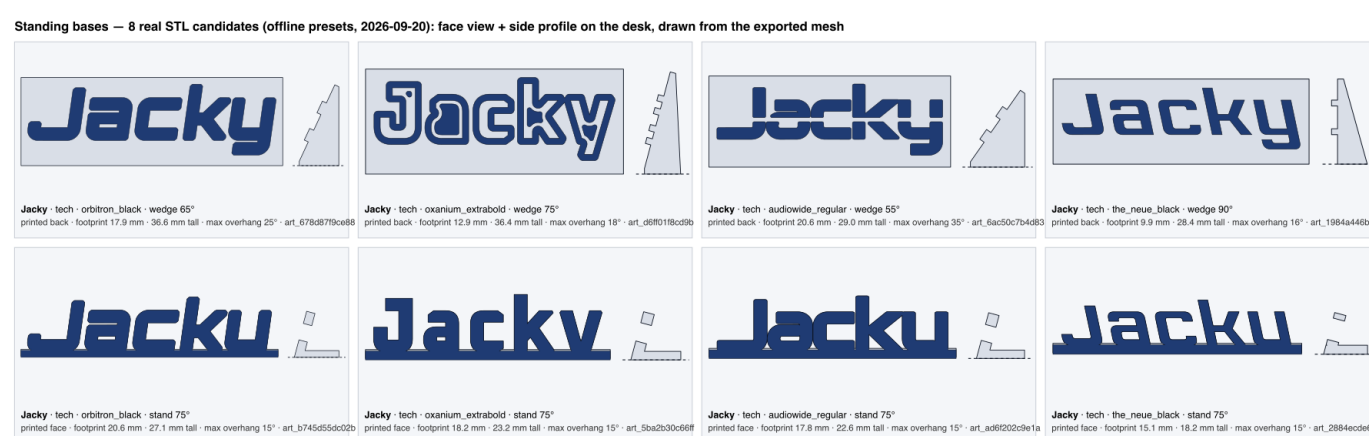


Figure 3. Standing Design Vector Graphics Preview

Validation & Repair

- Predictive Pre-Build Validation (Step 4):** Before CAD compilation, a design harness applies early planar feature checks. The AI model receives actionable feedback for structural failures, allowing up to two predictive revision rounds.
- Artifact Generation & Screening (Steps 5–6):** OpenSCAD executes the candidate builds, followed by rigorous evaluation through 15 geometry gates. Exported meshes undergo strict feature screening, including support and stability tests for standing pieces.
- Closed-Loop AI Repair Layer:** Deterministic validators identify parameter, geometry, and feature-level failures and pass the resulting failure reports to a specialized AI Repair Layer. The revised candidate is rebuilt and evaluated by the same validators. The loop is capped at two repair rounds, and unresolved failures remain explicitly reported.
- Full Auditability:** Every proposal, failure report, revision, and validation outcome is recorded in the artifact report and bound to the exported STL by SHA-256.

Results

An offline engineering benchmark of 40 requests produced 160 ready STL candidates. Of these, 137 passed directly, while 23 adopted repaired revisions; 20 candidates used measured stroke reinforcement. Median build time was 0.38 s per candidate. Separate v1.4 evidence produced 16 of 16 letterform-recipe STLs and 8 of 8 standing-base STLs without required-gate failures. In one minimal live Terra run, a 14.2 s model call produced four proposals and four engineering-review STLs. These datasets remain separate: the 160-candidate benchmark used offline pre-sets and did not run slicing.

Three selected artifacts were printed at the Duke Innovation Co-Lab on an Original Prusa CORE One using 0.4 mm PLA. All three prints completed successfully, with no visible bed detachment, gross deformation, or closed counters.

Evaluation Metric	Direct (w/o Repair)	Complete (w/ Repair)	Δ Improvement
Pre-Build Validation	88.8%	98.8%	+10.0%
STL Generation	97.5%	99.4%	+1.9%
Final Geometry Gates	85.0%	98.1%	+13.1%
First-Pass Yield (FPY)	81.9%	—	—
Overall Pipeline Yield	81.9%	97.5%	+15.6%

Table 1. Pipeline Reliability Evaluation and Ablation Results.



Figure 4. Result View

Future Work

Potential applications include personalized nameplates, event signage, classroom fabrication, and rapid prototyping of other text-based products. But the project is still in early stages, and several areas for improvement remain.

- Calibrate the current 1.2 mm feature threshold** through controlled prints across materials and printers.
- Confirm** production printer profiles and add per-feature G-code coverage checks
- Expand** live-model evaluation beyond a single Terra case, measuring diversity, latency, harness revisions, and repair success.
- Validate direct AI-drawn glyph paths** and conduct blinded user-preference studies.

References

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